AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 28 · STUDENT LESSON

Prompt Experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Compare matched cases. Inspect the failures. Choose by a declared rule.

Use complete paired records, explicit denominators and important-condition checks to evaluate prompt variants.

Keep one habit: Controlled setup → case-level outcomes → total and slice → declared selection → fresh final evidence.

MAIN LESSON · SECTION 1

Which prompt improves the results that matter?

> Compare matched cases and check important failures before choosing.

Your learning goals

I can explain the **mechanism** and terminology of controlled **prompt** experiments.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall outputs versus evidence

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Predictive vs Generative: a plausible continuation does not establish a current fact. A prompt comparison therefore needs a task contract, fixed **evidence** and explicit success criteria.

Your end goal

By the end, I can compare **prompt**s on **paired case**s and inspect a subgroup before choosing.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

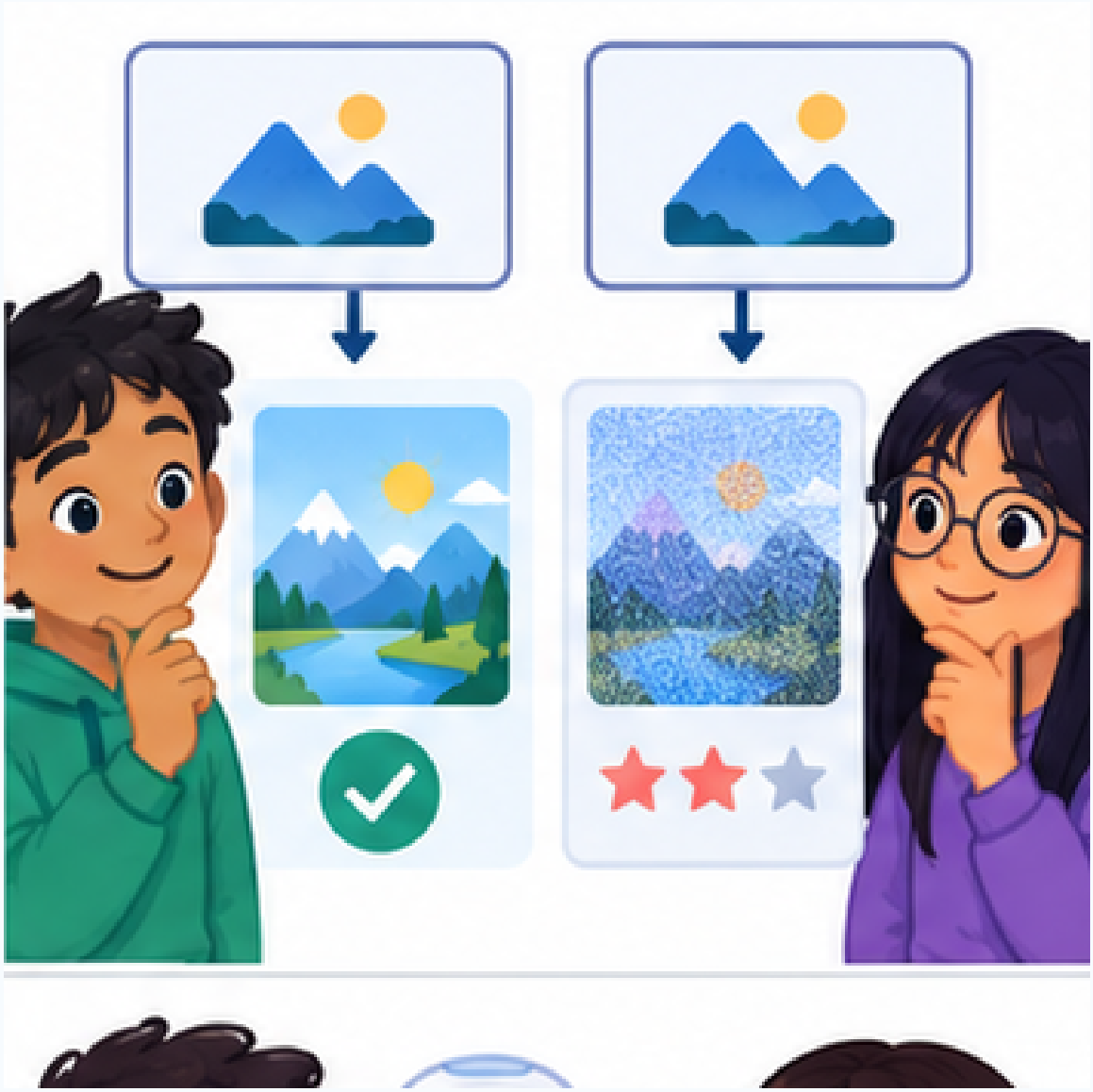
Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 28: **Prompt** Experiments

Prompt quality is evaluated by **controlled comparison**s and success criteria.



THINKER MISSION

Change one **prompt** variable at a time and score results.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Set the question before looking at results

> A fair comparison needs the same cases, evidence and scoring rules.

The controlled-comparison contract

A **prompt** experiment compares complete specifications using the same task cases. Define success before inspecting **output**s; change one intended instruction and preserve other settings.

Distinguish supplied classroom records from live results

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

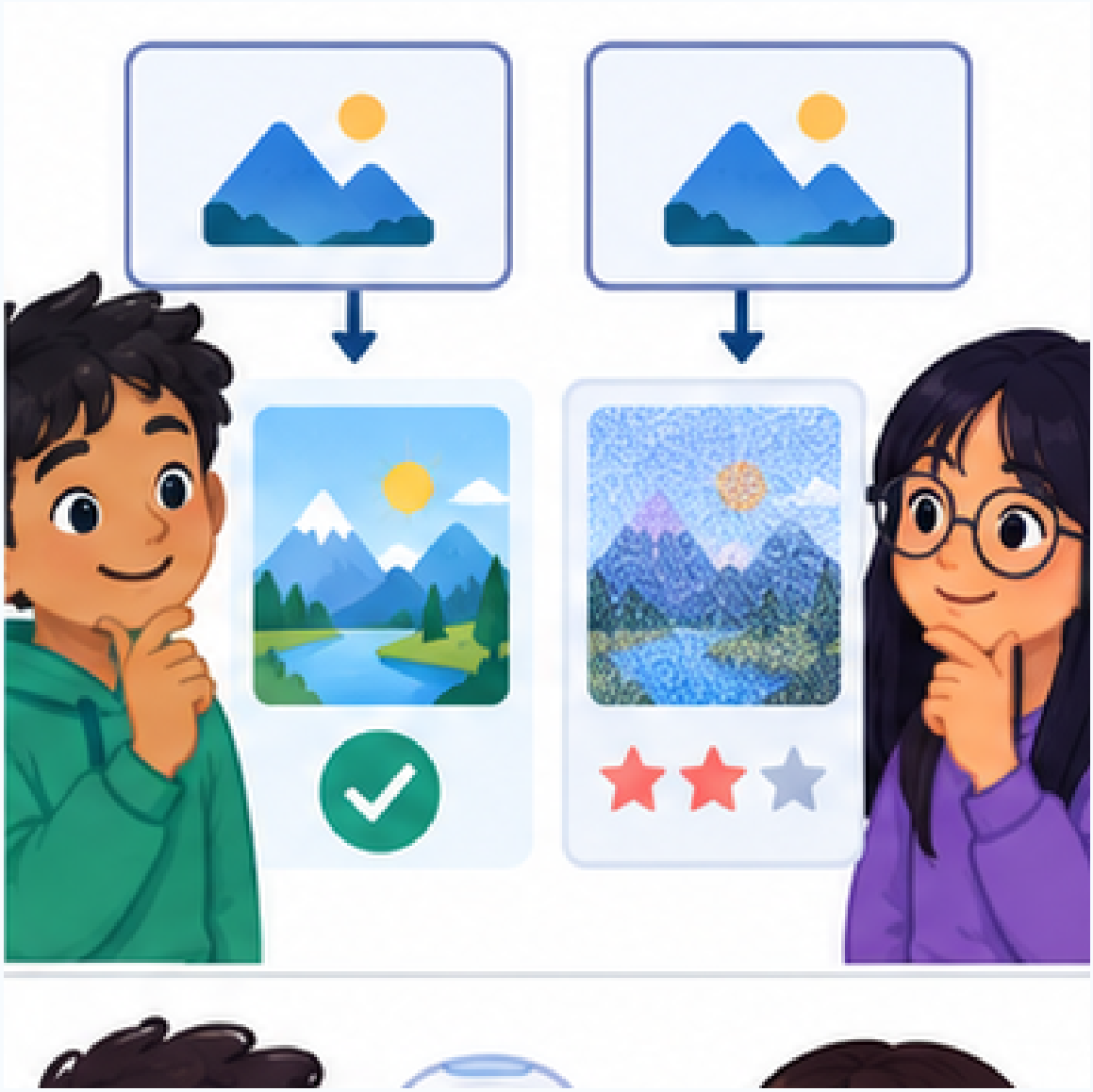
These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **prompt** experiments to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Trace the six-case development comparison

> Overall gains and subgroup gains are different questions.

Keep evaluation conditions fixed

Evaluate both **prompt**s on **paired case**s with fixed model/version, source records, **output** contract and scoring **rubric**. Use development results for selection, then reserve new final cases.

Score each case rather than counting attractive sentences.

Read the complete six-case record

Six-case development record	P	Q
Pass IDs	1,2,3,4	1,2,3,5,6
Total passes /6	4/6	5/6
Missing-approval IDs4,6	case4 passes; case6 fails	case4 fails; case6 passes

Check denominators and keep the raw record

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain what a picture label represents

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

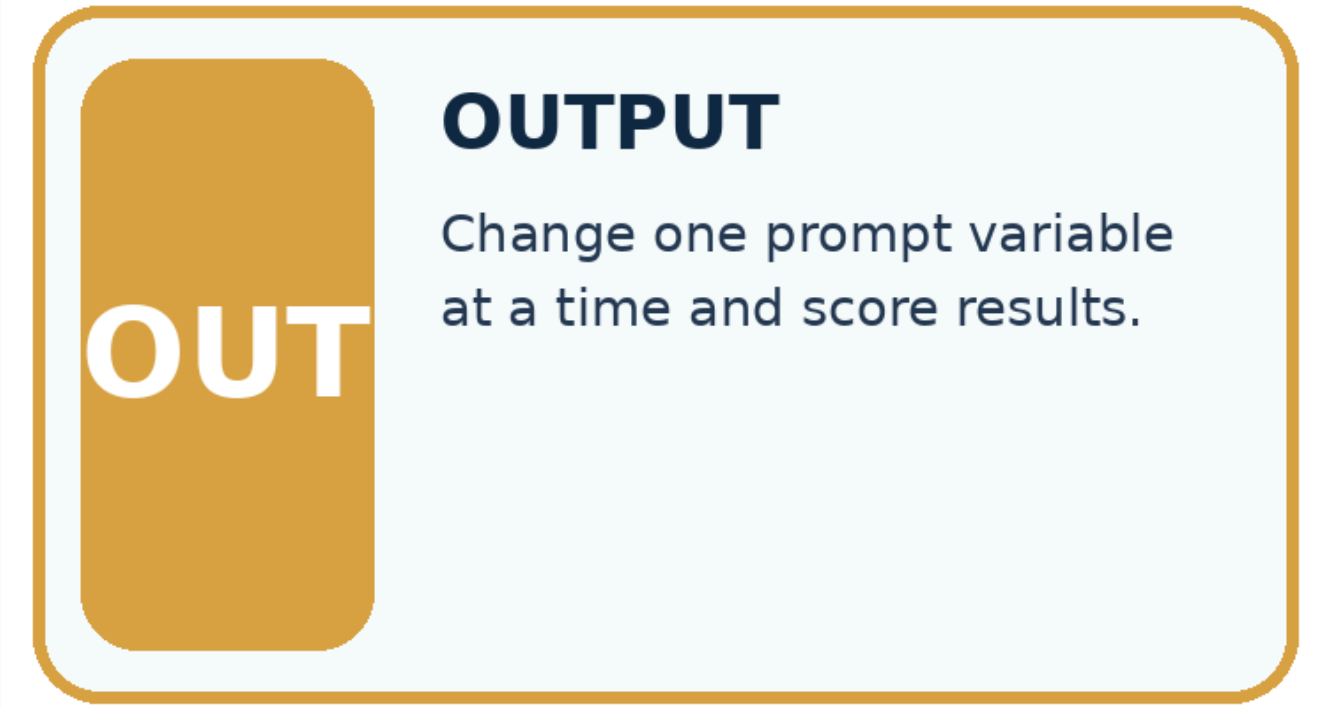
VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Change one prompt variable at a time and score results.
↓
↻
PROCESS
Prompt quality is evaluated by controlled comparisons and success criteria.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Controlled prompt experiments — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.	Evaluate both prompts on paired cases with fixed model/version, source records, output contract and scoring rubric.	P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

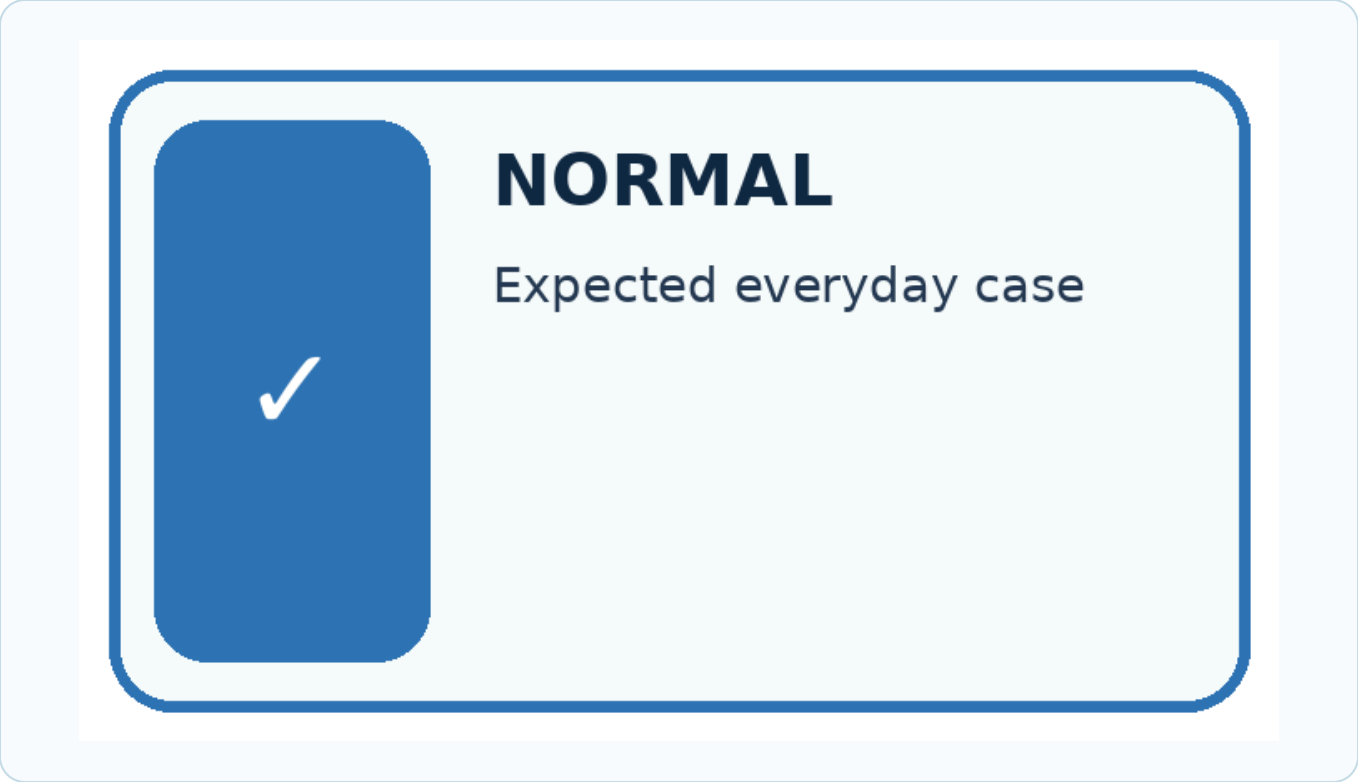
Connect the picture and the explanation

MAIN LESSON · SECTION 4

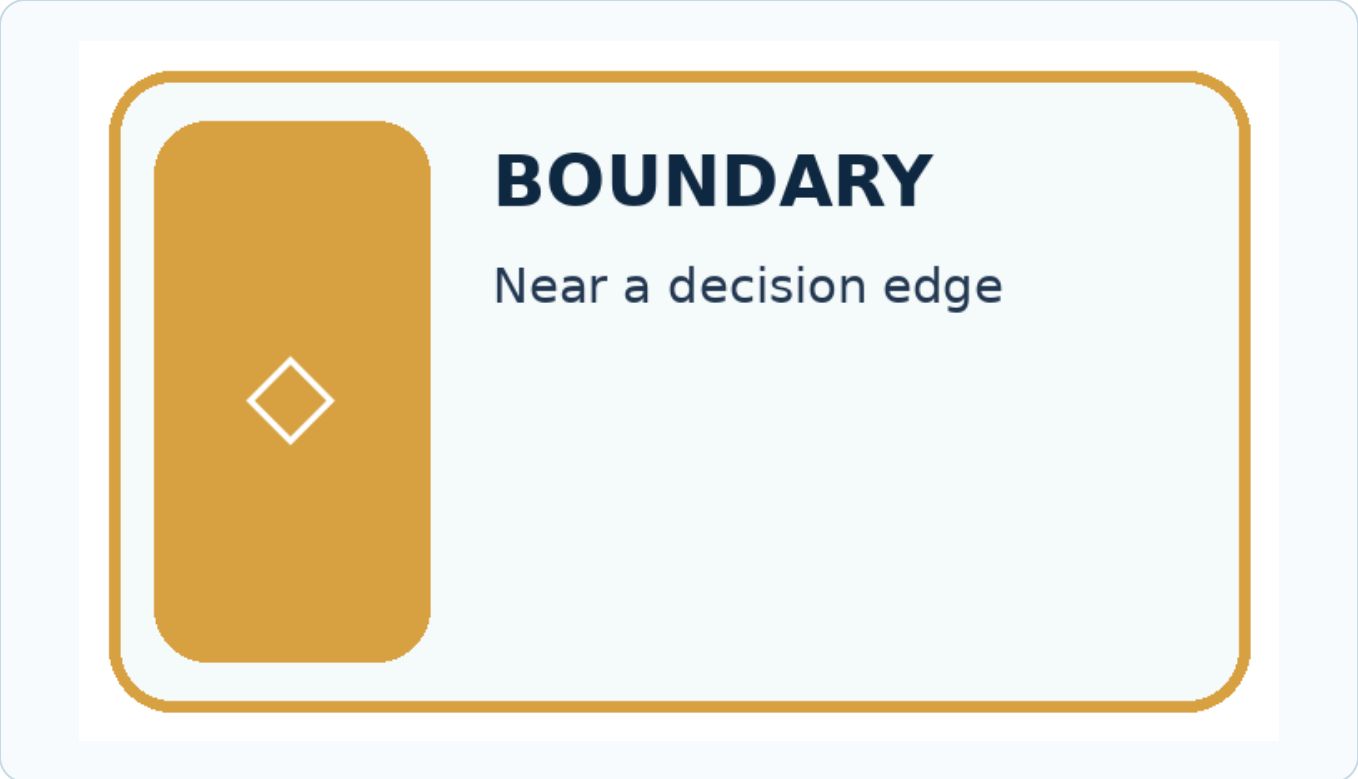
Compare prompt tests by the evidence they preserve

> A different question set confounds the comparison.

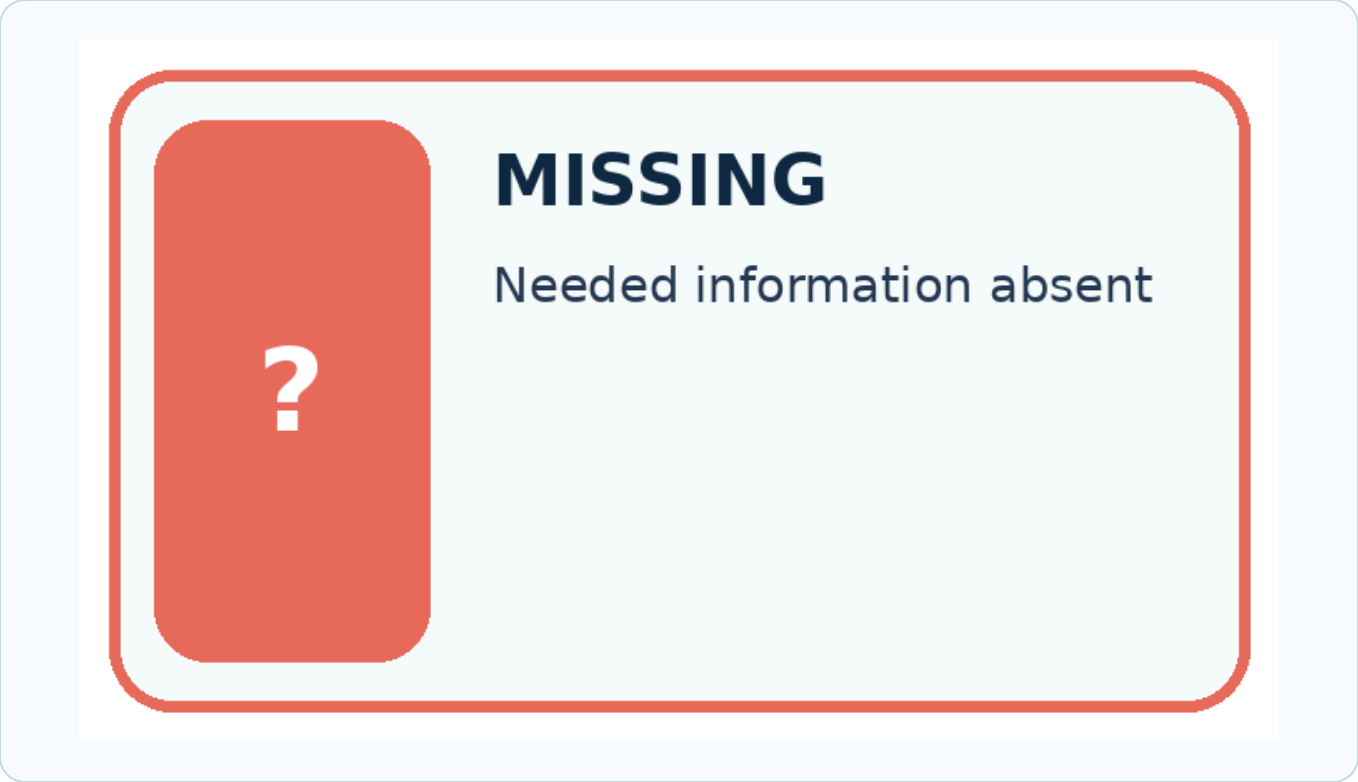
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Make missing-source correctness a required selection gate.

▼ Check after predicting

Prediction: Neither **prompt** clears a 2/2 gate on these records.

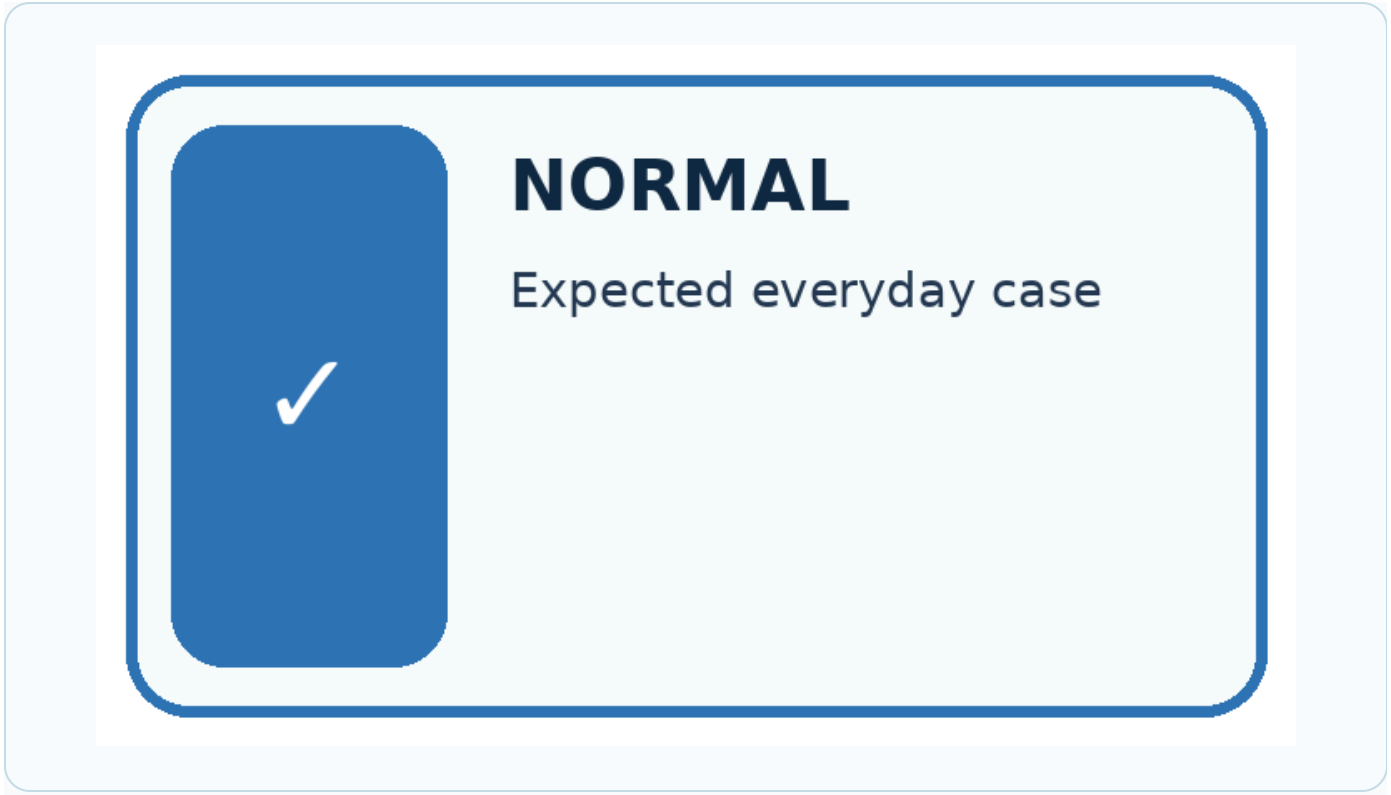
Reason: P fails 7 and Q fails both 4 and 7; both need revision.

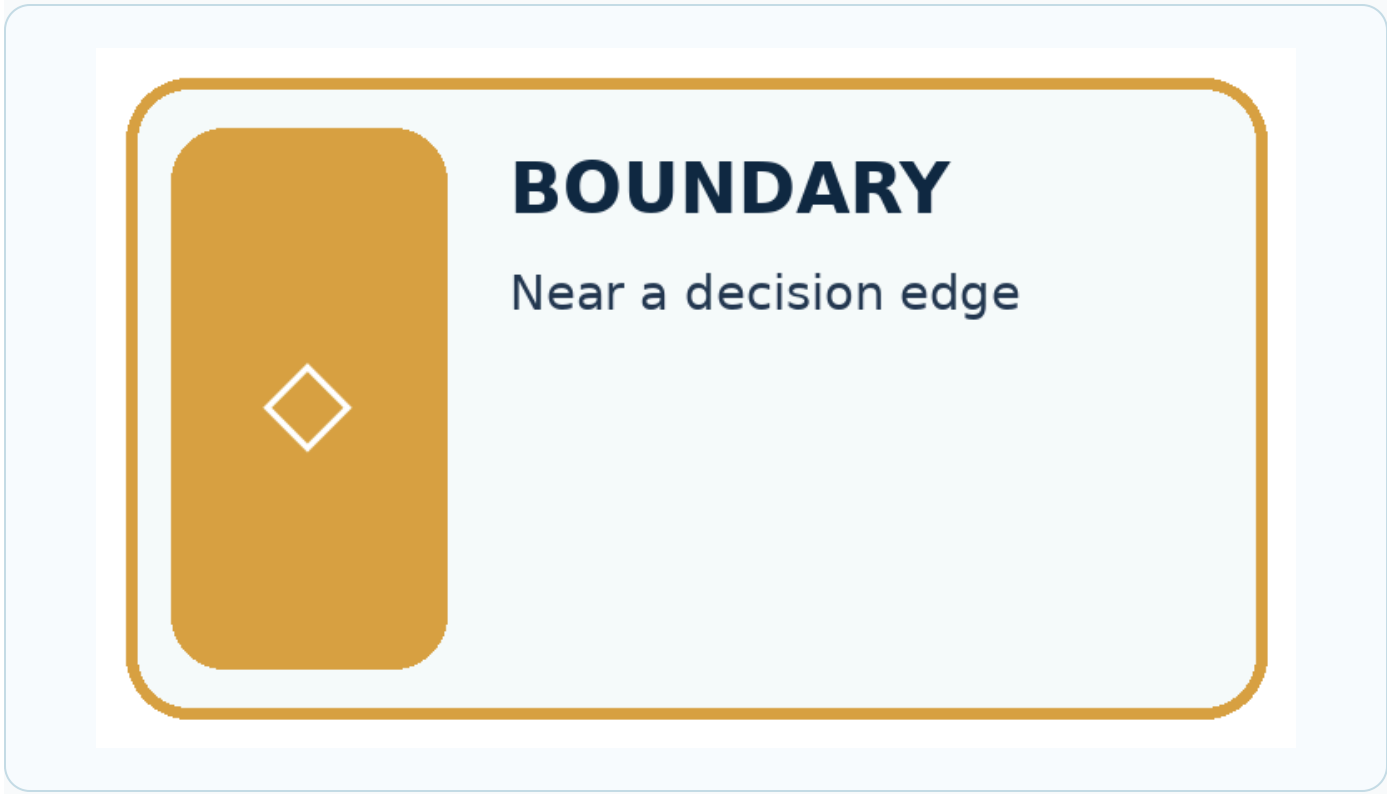
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

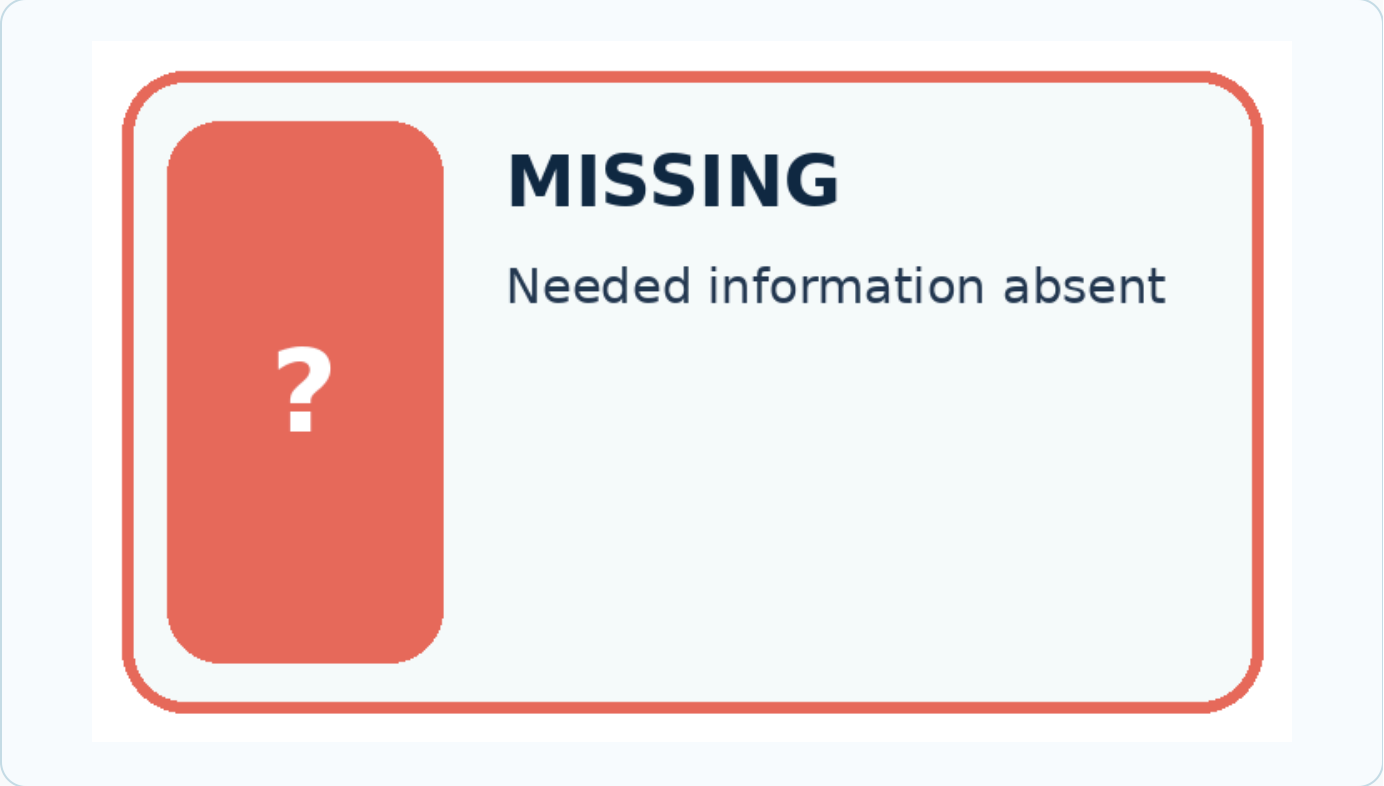
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Compare six-case results, then a separate seven-case picture

> The missing-approval slice and missing-source slice belong to different datasets.

Read the six-case counts and interpretation

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

Start the separate seven-case book-help dataset

A fictional book-help project tests two **prompt** versions on seven shared development cases.

P passes cases 1,2,3,4. Q passes 1,2,3,5,6. Cases 4 and 7 involve missing sources.

Same **input**s, system configuration and **rubric**; only the **prompt** changes. These are invented records.

Trace overall passes and missing-source failures



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Paired setup	Use the same seven cases and pass criteria for both prompts.
2. Overall rates	P=4/7; Q=5/7.
3. Missing-source slice	P passes 1/2; Q passes 0/2 in cases 4 and 7.
4. Decide explicitly	Q improves overall count but worsens missing-source behavior; use the declared priorities.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Explain the selection trade-off

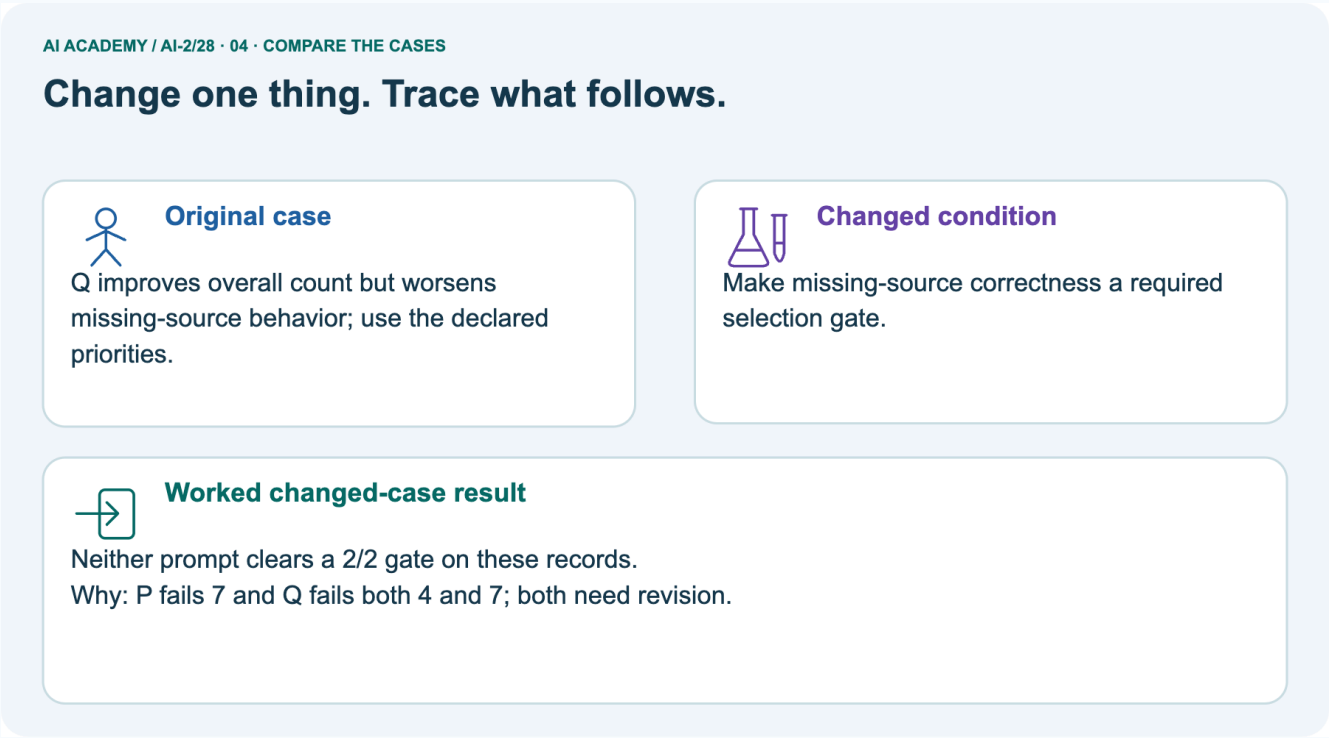
A larger total can conceal a more serious failure category. Paired records show which cases changed, not just how many passed. If refusing unsupported claims is mandatory, Q's overall advantage may not justify selection.

Keep the failing examples for development and obtain fresh cases after choosing a final version.

Predict a strict missing-source gate

- Change one thing: Make missing-source correctness a required selection gate. Predict the result before checking the explanation. What remains the same?

▼ Check after your prediction



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the limit on the claim

- Seven development cases cannot establish causal learning benefits or untouched final performance; repeated generation may also vary.

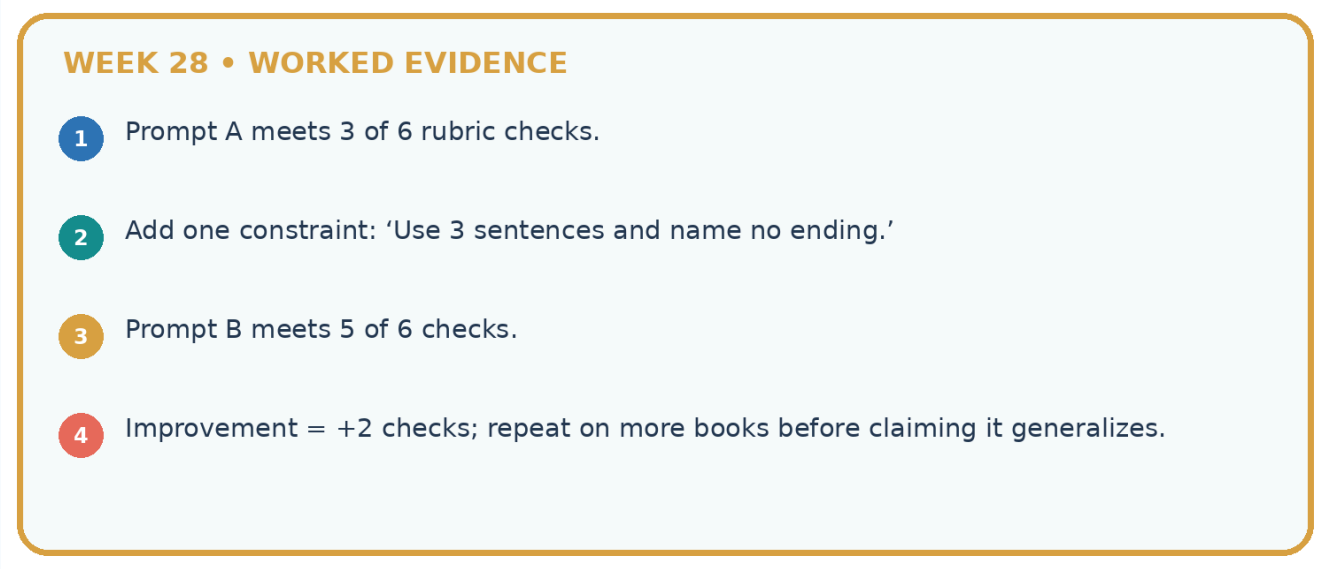
- Read the labelled picture: Name the case P passes and Q fails, and explain why it matters.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. **Prompt** A meets 3 of 6 **rubric** checks.

2. Add one **constraint**: 'Use 3 sentences and name no ending.'

3. **Prompt** B meets 5 of 6 checks.

4. Improvement = +2 checks; repeat on more books before claiming it generalizes.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

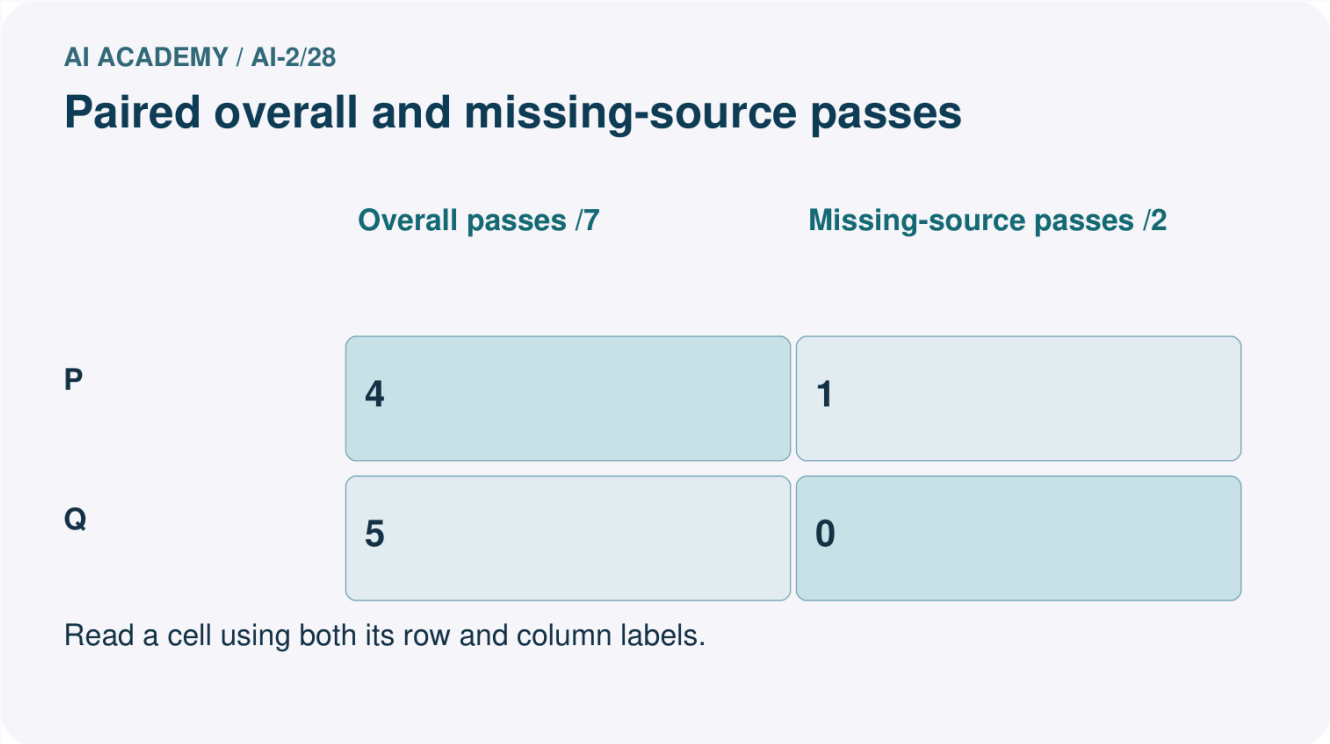
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Prompt Experiments

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/28

Prompt Experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Paired setup

Use the same seven cases and pass criteria for both prompts.



2. Overall rates

P=4/7; Q=5/7.



3. Missing-source slice

P passes 1/2; Q passes 0/2 in cases 4 and 7.



4. Decide explicitly

Q improves overall count but worsens missing-source behavior; use the declared priorities.

CHANGE ONE THING

Make missing-source correctness a required selection gate.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the failure hidden by a larger total

> Compare case-level changes before accepting an aggregate claim.

Repair the claim using the seven-case record

Claim to repair: Q passes more cases, so it is safer on missing sources. Use the seven-case picture: Q passes5/7 overall but0/2 missing-source cases; P passes4/7 overall and1/2 of that slice. Q also loses case4 that P passed. Neither clears a required2/2 missing-source gate. A larger total alone cannot justify the safety claim.

State a test that could change your decision

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: One impressive output proves a prompt is reliable. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise the six-case trace with less help

> Keep the seven-case picture and independent eight-case task separate.

The development-versus-final boundary

Prompt selection on development cases cannot be reported as independent final performance.

Complete the six-case W2 trace

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Controlled **prompt** experiments”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Challenge an overgeneralised conclusion

- Use the worked case for Controlled **prompt** experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the separate seven-case regression

- Name the case P passes and Q fails, and explain why it matters.

Explain the picture

> **My explanation using a pictured step and the given facts**

Keep independent evidence separate

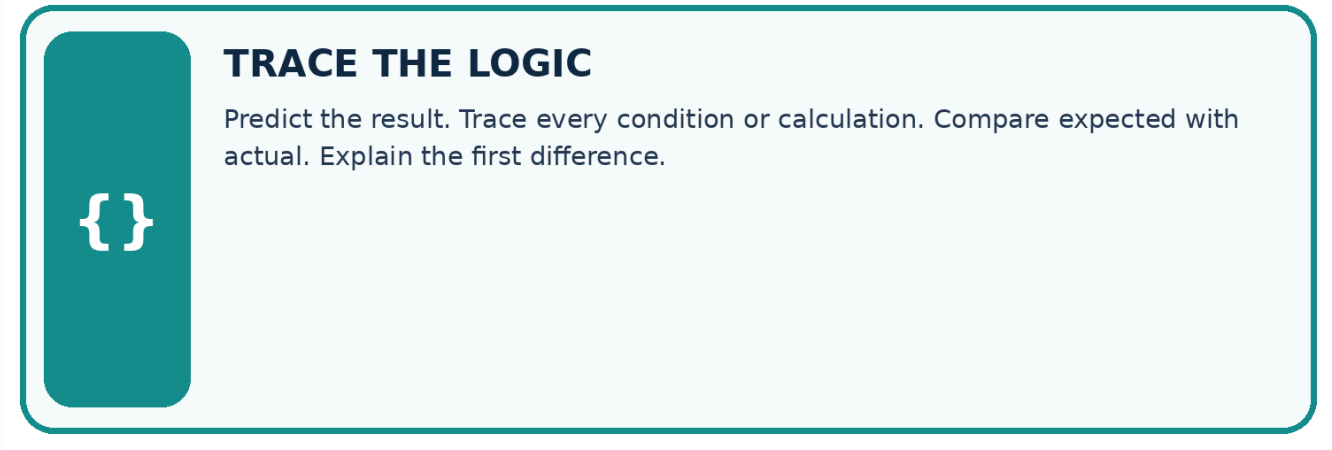
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Audit a complete paired-results record

> All seven outcomes are supplied; report what they establish and what remains unmeasured.

Materials, procedure and scope

1. Paper investigation using complete supplied records: use IDs1–7; P pass flags are [1,1,1,1,0,0,0], Q flags [1,1,1,0,1,1,0]. IDs4 and7 are missing-source cases. First predict which prompt has more passes and which does better on that slice. Count total passes, shared passes, gains and losses, and slice passes. Declare before selection either total-pass priority or a mandatory2/2 missing-source gate; compare the two decisions. These are invented scored outcomes, not outputs generated in this classroom. The actual prompt strings, model/version and rubric are not supplied, so these records support a paired outcome audit, not reproducing an actual prompt experiment or attributing a gain to one exact wording change. To run a new experiment, freeze those missing specifications, use the same cases, predeclare a rubric and repeated-run plan if outputs vary. Reserve the eight-case task for independent work.

Keep prediction, trace and decision

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

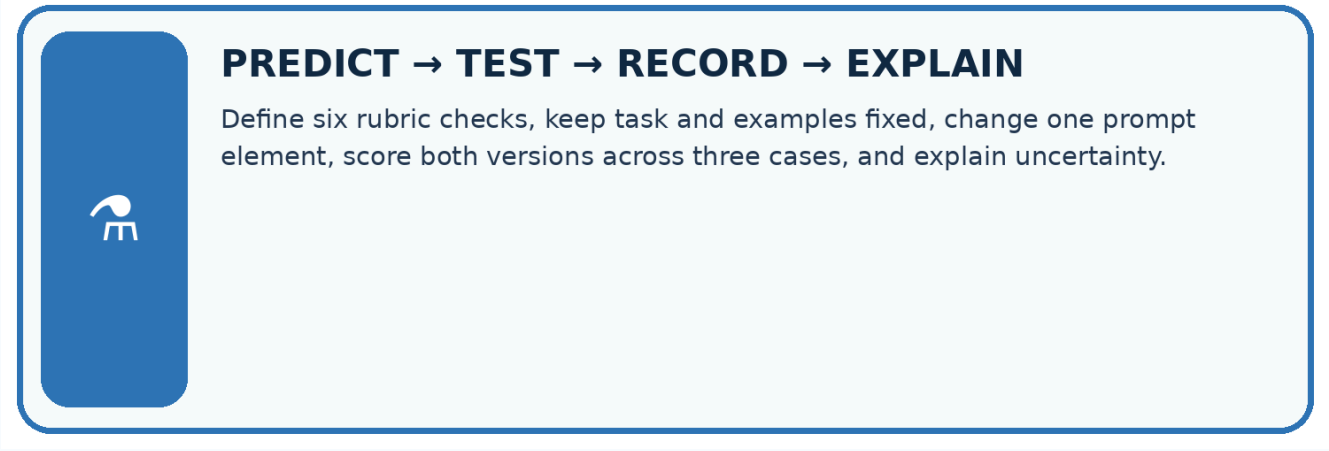
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Audit the supplied seven-case pass flags as a paper investigation. Use fixed denominators and explicitly declared selection priorities. No live **prompt** outputs, actual **prompt** strings or repeated-run results are supplied.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare the fresh eight-case results

> Use the stated total-pass selection rule and separately report the special-case regression.

The independent case

On eight **paired case**s A passes 1–5; B passes 1–4, 6, 7. Special cases are 5 and 8. Compare overall and special-case results; select by total passes.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Repair only after feedback

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply a new case independently

Week 28 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Prompt A and B are tested on different questions. B scores higher. Can wording alone explain the difference? Design a **controlled comparison**.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Test generated book explanations with a fixed **rubric** and prohibit invented details.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Prompt A is tested on easy requests and B on requests with missing data. B scores lower. Design a fair comparison and specify what to do when **output**s vary across runs.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep a reproducible book-explanation comparison

> A prompt variant is a proposal until its outputs have been checked.

This week's concrete project contribution

For book explanations, freeze catalogue version, reader requests, prompt versions, output format and the scoring rubric before comparison. Require factual claims to be supported by supplied records, and specify handling of missing sources. Keep per-case failures and an important-condition slice beside the total. Select using development cases, then use untouched final cases; if outputs vary, prespecify repeated runs and report their spread. Until real outputs are collected, label the work a plan or invented-record audit.

Run controlled recommender explanation experiments with **paired case**s and report important failure conditions separately.

Keep the checkpoint and human choice

THIS WEEK'S CHECKPOINT Test generated book explanations with a fixed **rubric** and prohibit invented details.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

> **Write or draw your reasoning and evidence.**

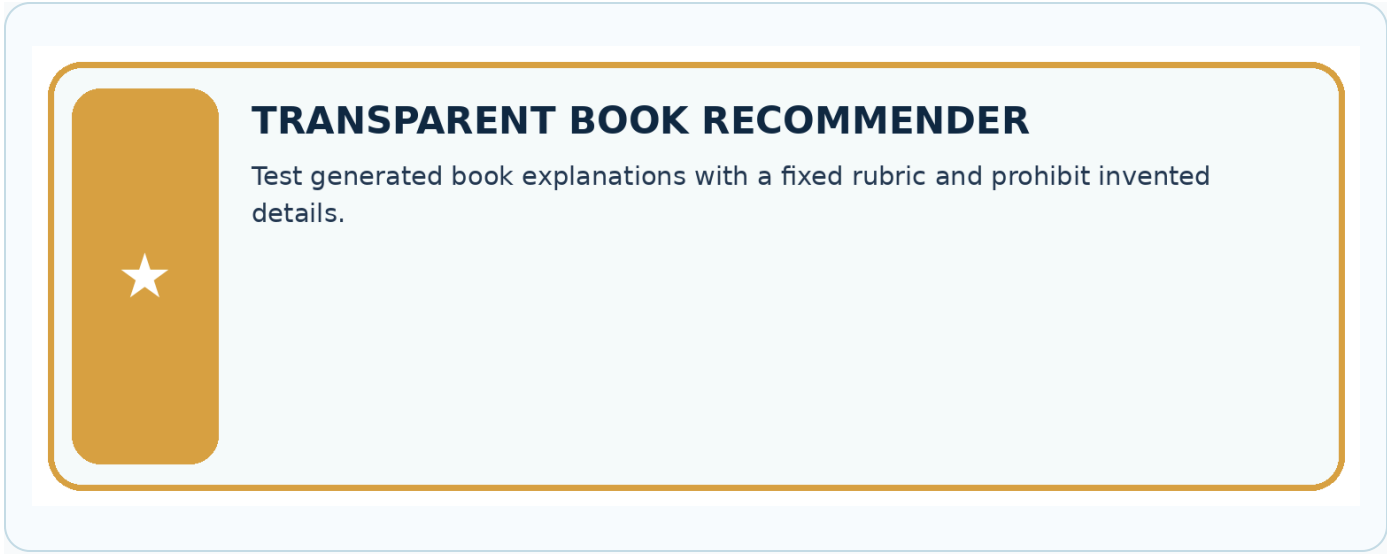
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: show the total, the slice and the evidence boundary

> A development choice is not an untouched final result.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of controlled **prompt** experiments

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
prompt	Input instructions and context supplied to a generative system.	
constraint	A condition that a valid solution must satisfy.	
criterion	One stated standard used to judge a result.	
controlled comparison	A comparison designed to keep relevant conditions alike except for the factor being studied.	

Terms for the connected explanation

Word	Meaning
Paired case	The same case evaluated under each compared procedure
Rubric	Prespecified criteria for judging an output

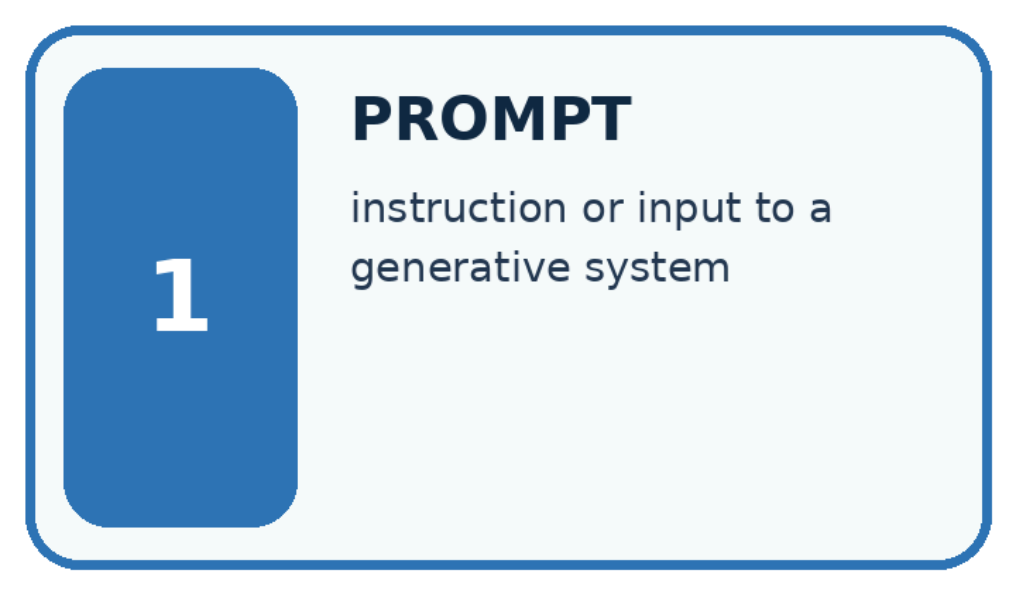
Word	Meaning
Regression	A previously passing condition that fails after a change

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<summary>Earlier wording, alternatives and picture transcripts</summary>

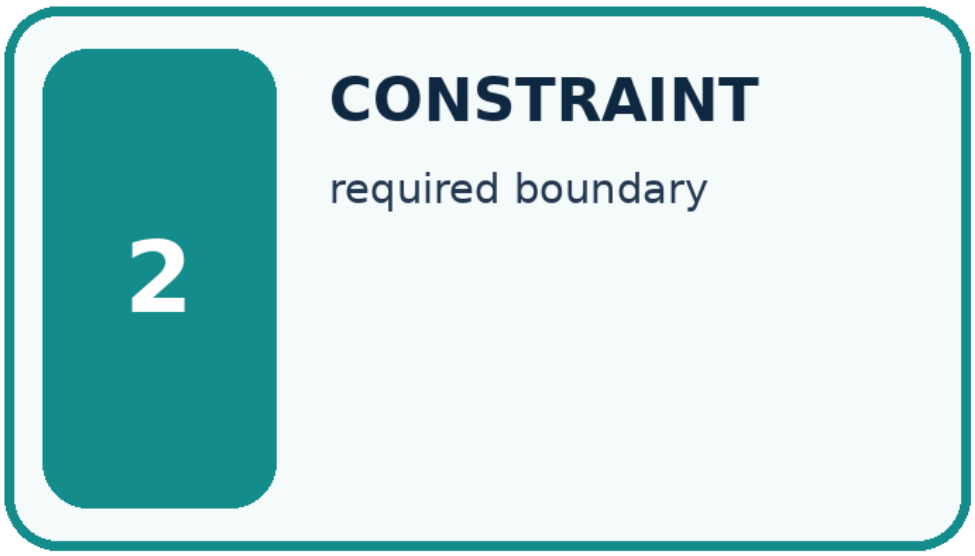
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



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2
CONSTRAINT
required boundary


3
CRITERION
rule for judging success



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

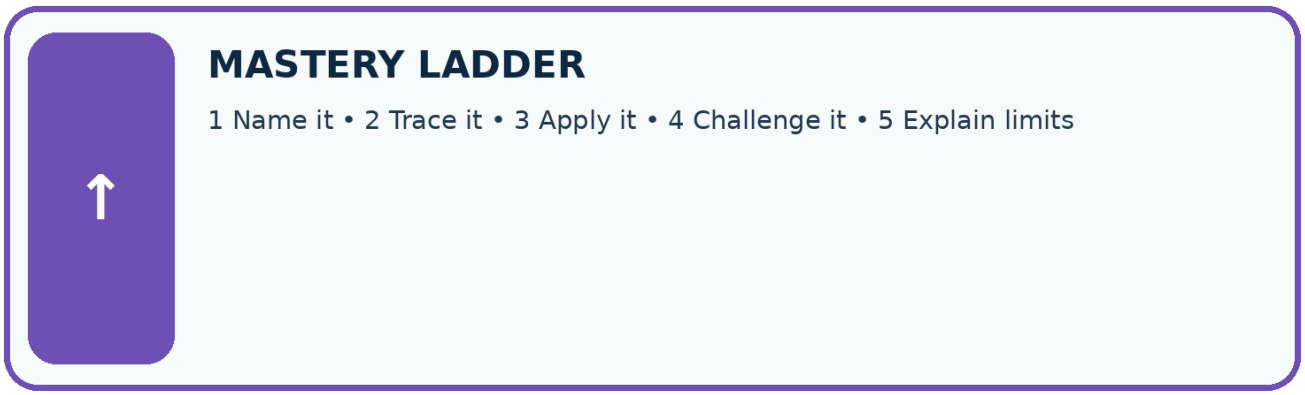
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Compare a new paired record independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

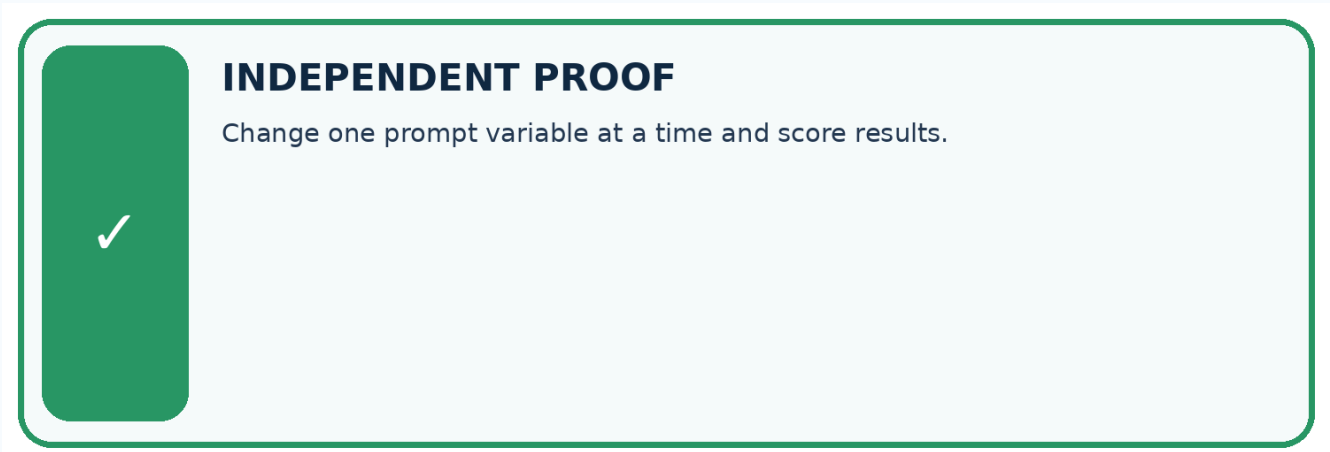
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Compare a new paired record independently is the selected independent case.



EXPLANATION CHECK Count total passes and the named slice separately; use the stated priority and qualify development evidence.

Write or draw your reasoning and evidence.

▼ **Extra practice: Compare prompts under the same conditions**

REFERENCE · SECTION 13

Extra practice: Compare prompts under the same conditions

> Try the question before opening its worked explanation.

Read the supplied case and question

Prompt B adds under 100 pages and explain the match to **Prompt** A. Both use a catalogue.

Design a fair comparison.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Use identical requests, catalogue and scoring criteria. Check **constraint** compliance and supported explanation; repeat stochastic **output**s when relevant.

Evidence or stage	Value or result
Prompt A	Original catalogue request
Prompt B	Add under 100 pages and explain the match
Hold fixed	Catalogue, requests and scoring
Check	Constraint compliance and supported explanation

▼ Earlier wording, alternatives and picture transcripts

Week 28 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Compare prompts under the same conditions

Prompt B adds under 100 pages and explain the match to **Prompt** A. Both use a catalogue.

Design a fair comparison.

Use identical requests, catalogue and scoring criteria. Check **constraint** compliance and supported explanation; repeat stochastic **output**s when relevant.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of controlled prompt experiments	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional book-help project tests two **prompt** versions on seven shared development cases.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional book-help project tests two **prompt** versions on seven shared development cases.
- Which labelled step matches this detail? Q improves overall count but worsens missing-source behavior; use the declared priorities.
- 1. Decide explicitly
- 2. Paired setup
- 3. Missing-source slice
- 4. Overall rates
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Decide explicitly Q improves overall count but worsens missing-source behavior; use the declared priorities.

A larger total can conceal a more serious failure category. Paired records show which cases changed, not just how many passed. If refusing unsupported claims is mandatory, Q's overall advantage may not justify selection.

Keep the failing examples for development and obtain fresh cases after choosing a final version.

Changed-case question

- Changed-case practice: Make missing-source correctness a required selection gate. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Neither **prompt** clears a 2/2 gate on these records.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Neither **prompt** clears a 2/2 gate on these records. P fails 7 and Q fails both 4 and 7; both need revision.

Repair and save your work

Review Predictive vs Generative (ai-2/27). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Does matching the physical box in the changed table prove the generator measures boxes?. Return to this week and explain what you changed.

Review Text as Pieces (ai-2/25). Goal: Encode and decode text under an explicit toy vocabulary. Short practice: Why is the unknown ! at position 10 rather than position 9?. Return to this week and explain what you changed.

Review Nearest Neighbor Reasoning (ai-2/22). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why do k=1 and k=3 disagree for the original query?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

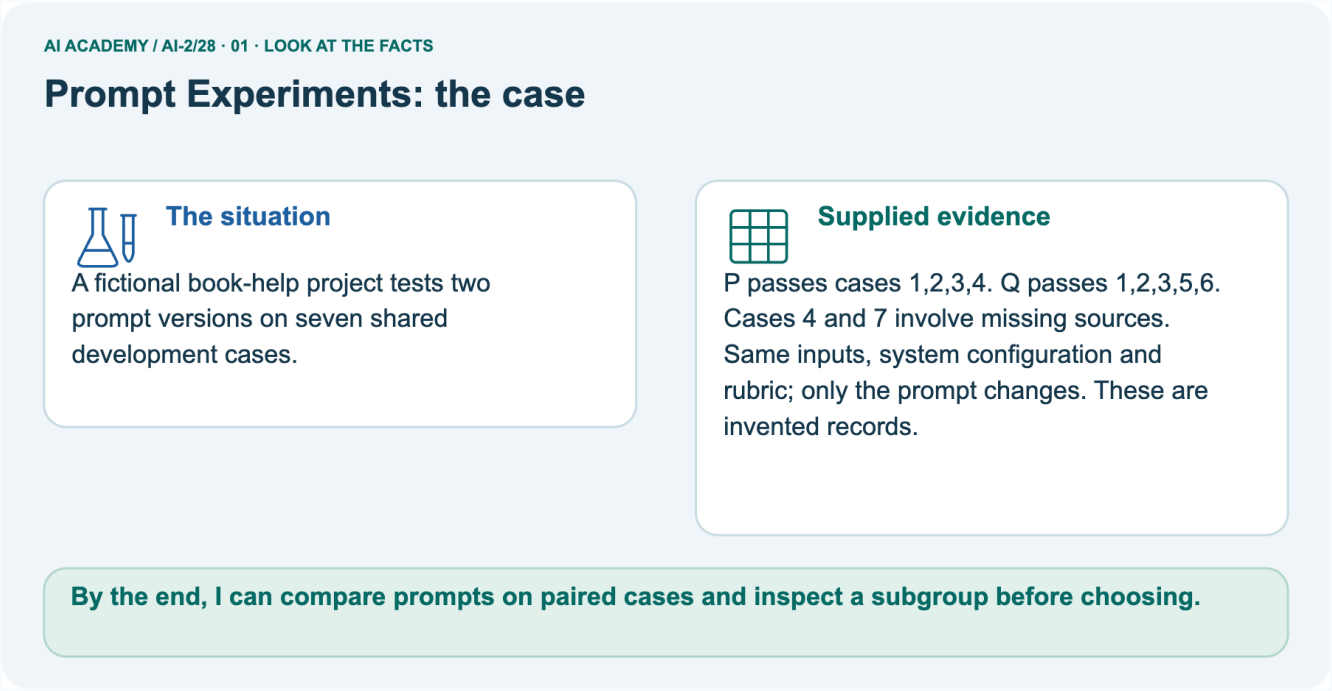
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Prompt Experiments: the case



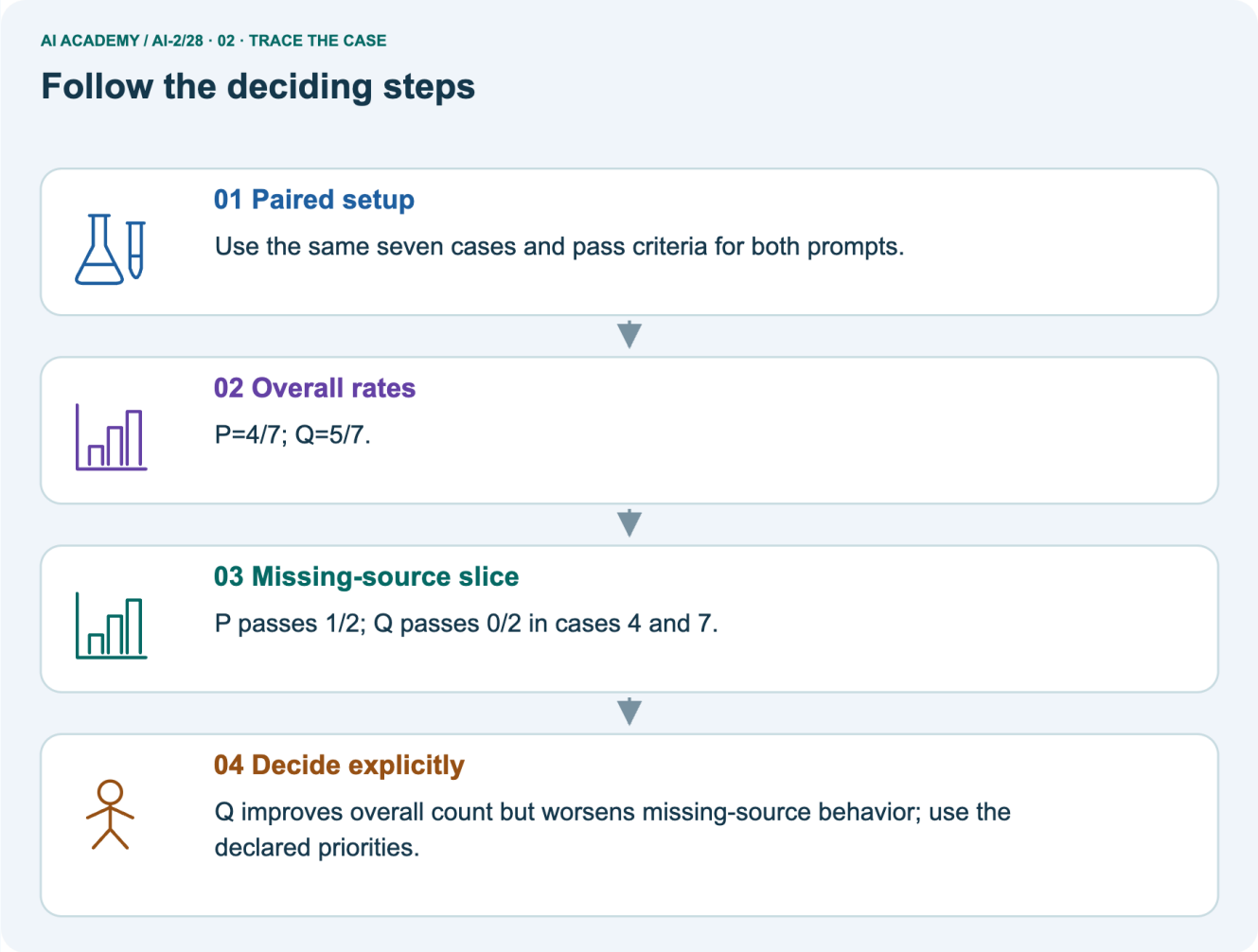
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional book-help project tests two **prompt** versions on seven shared development cases. Supplied facts: P passes cases 1,2,3,4. Q passes 1,2,3,5,6. Cases 4 and 7 involve missing sources.

Same **input**s, system configuration and **rubric**; only the **prompt** changes. These are invented records.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

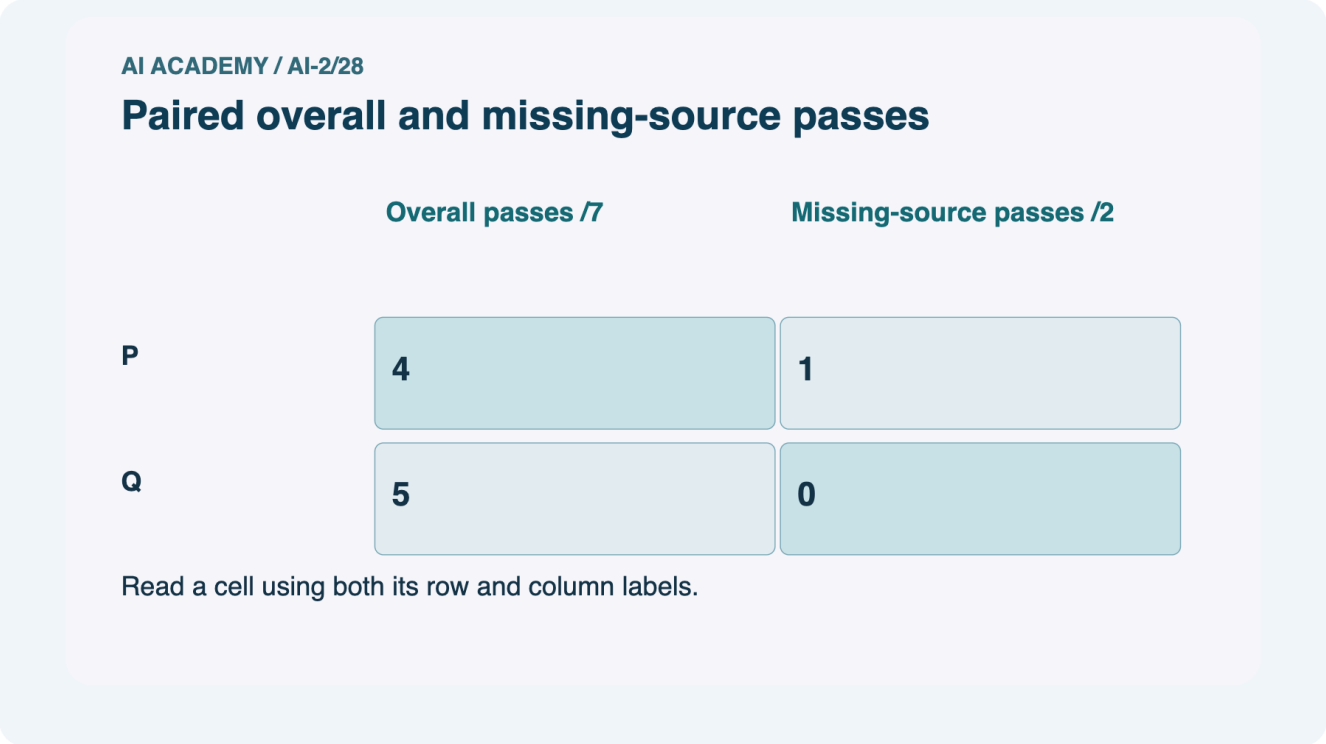


Follow the labelled picture with the matching explanation. Open the vector diagram

Paired setup: Use the same seven cases and pass criteria for both **prompt**s. → Overall rates: P=4/7; Q=5/7. → Missing-source slice: P passes 1/2; Q passes 0/2 in cases 4 and 7. → Decide explicitly: Q improves overall count but worsens missing-source behavior; use the declared priorities.

Point to each numbered stage and explain which recorded **input** it uses.

03. Paired overall and missing-source passes

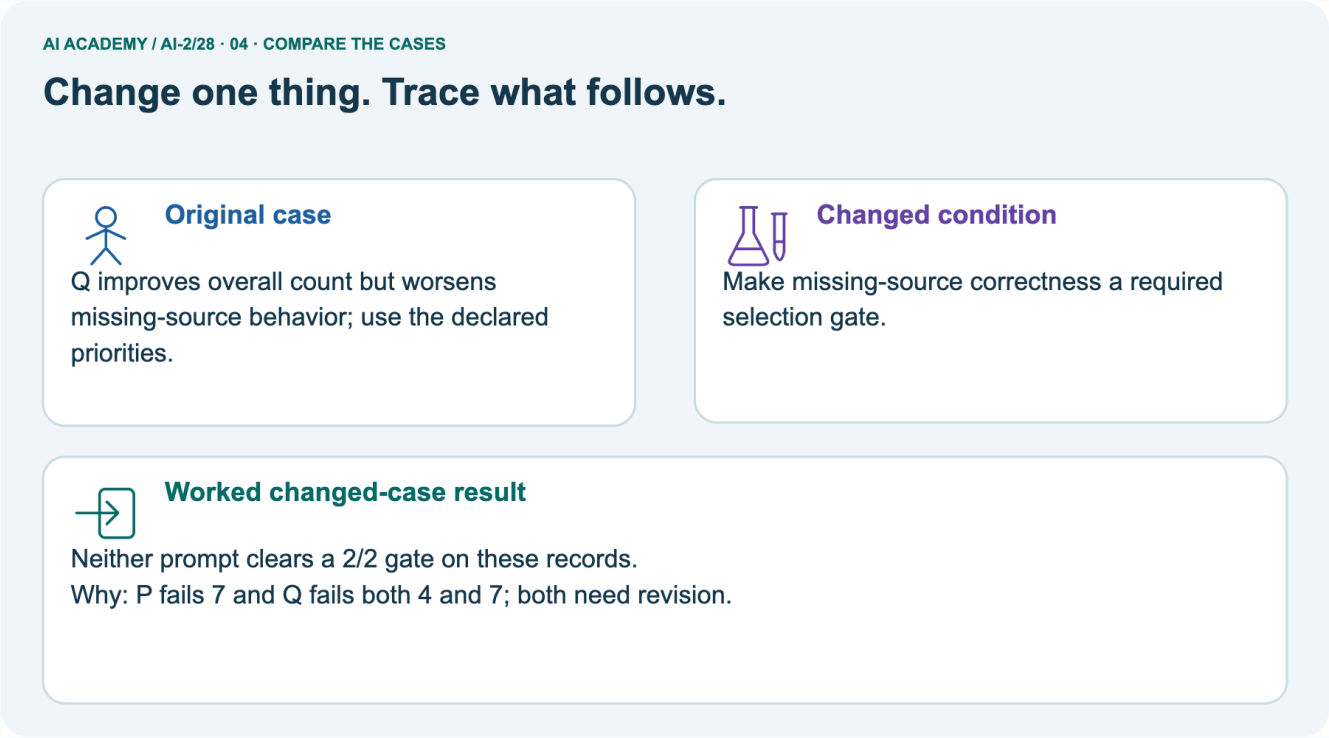


Follow the labelled picture with the matching explanation. Open the vector diagram

Paired overall and missing-source passes. P: Overall passes /7 = 4, Missing-source passes /2 = 1. Q: Overall passes /7 = 5, Missing-source passes /2 = 0.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



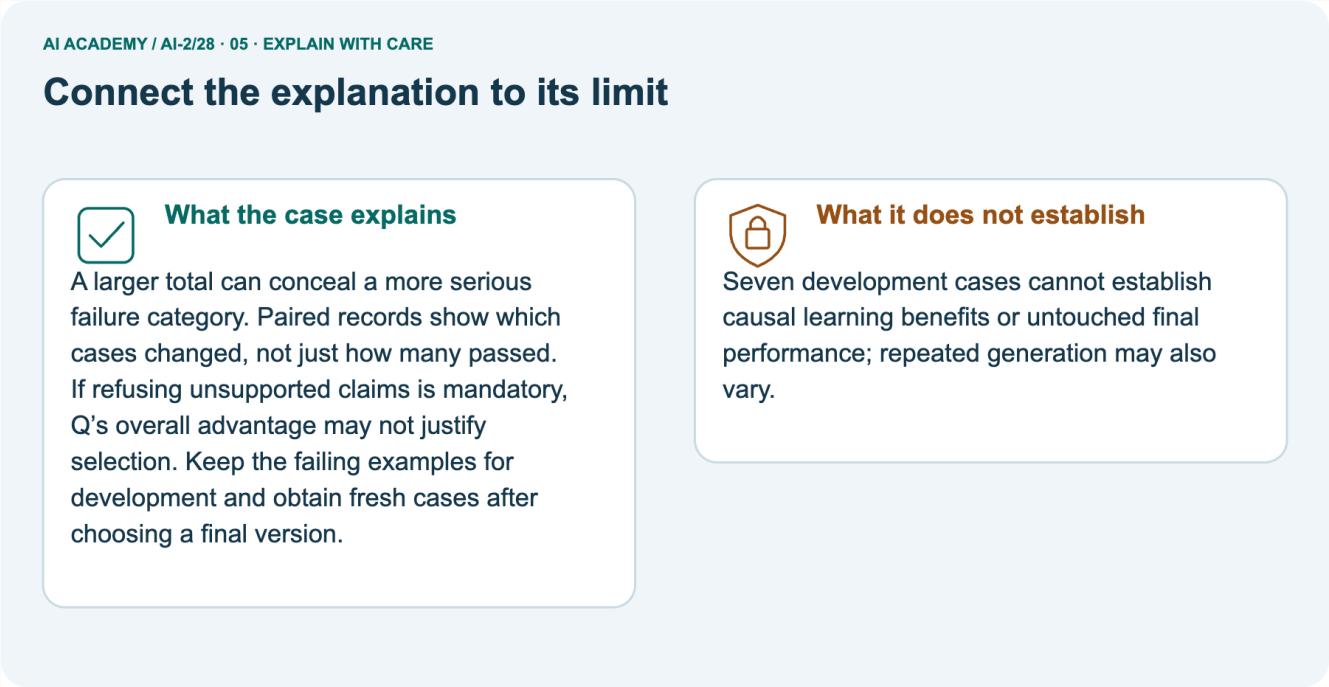
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Q improves overall count but worsens missing-source behavior; use the declared priorities. Changed condition: Make missing-source correctness a required selection gate.

Expected result: Neither **prompt** clears a 2/2 gate on these records. Reason: P fails 7 and Q fails both 4 and 7; both need revision.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

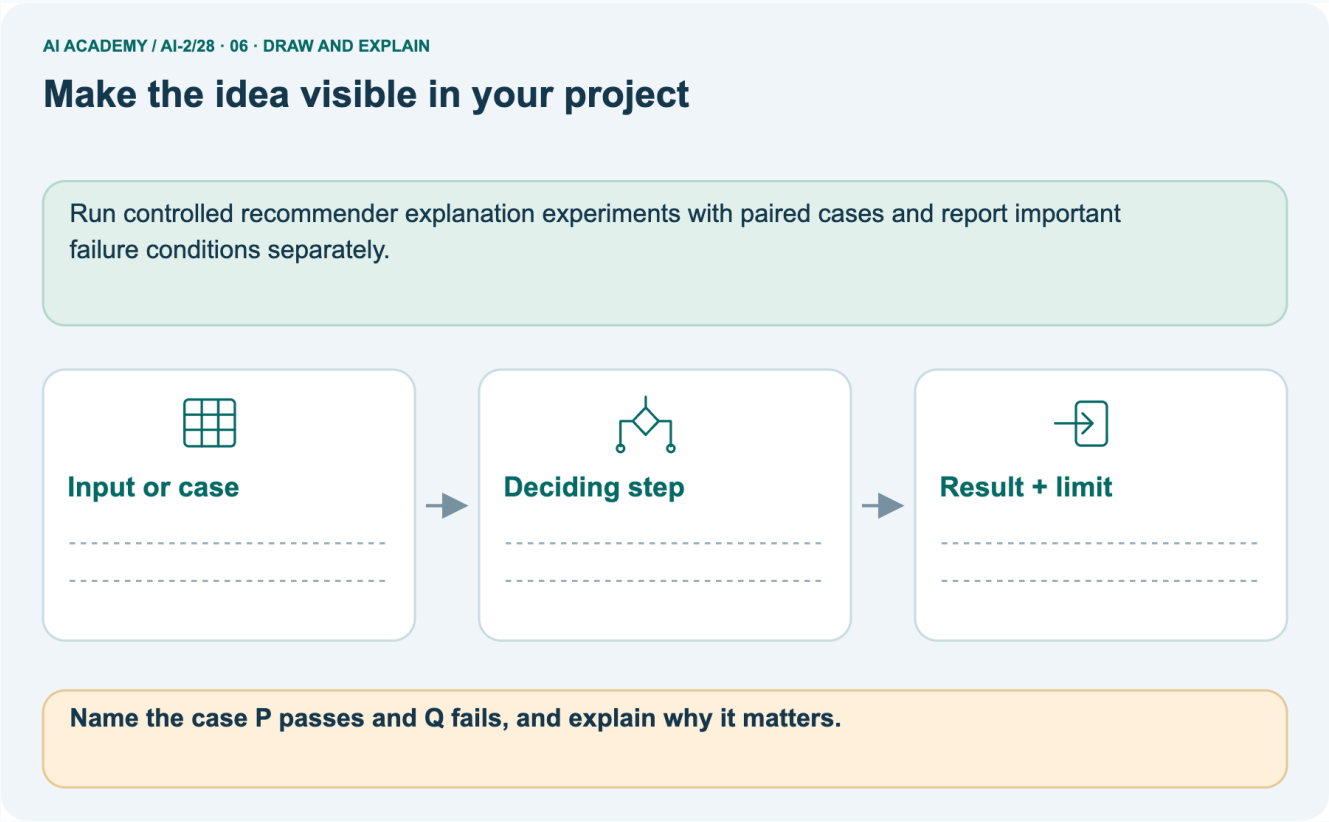
A larger total can conceal a more serious failure category. Paired records show which cases changed, not just how many passed. If refusing unsupported claims is mandatory, Q's overall advantage may not justify selection.

Keep the failing examples for development and obtain fresh cases after choosing a final version.

Limit: Seven development cases cannot establish causal learning benefits or untouched final performance; repeated generation may also vary.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Run controlled recommender explanation experiments with **paired case**s and report important failure conditions separately. Draw your **input**, deciding step, result and limit.

Supported question: Name the case P passes and Q fails, and explain why it matters.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Prompt Experiments: the case

02. Follow the deciding steps

03. Paired overall and missing-source passes

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.